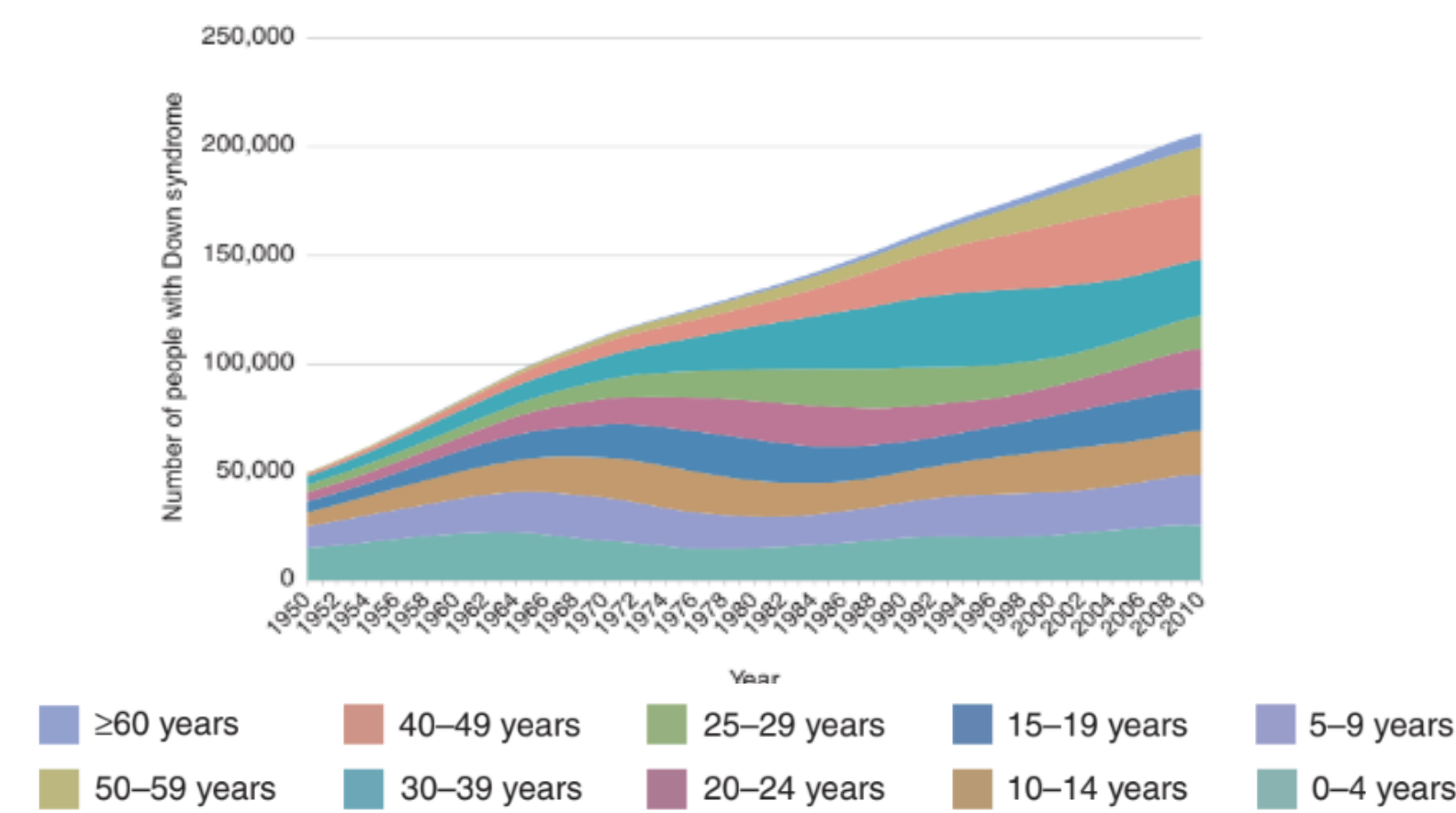


1. ABSTRACT

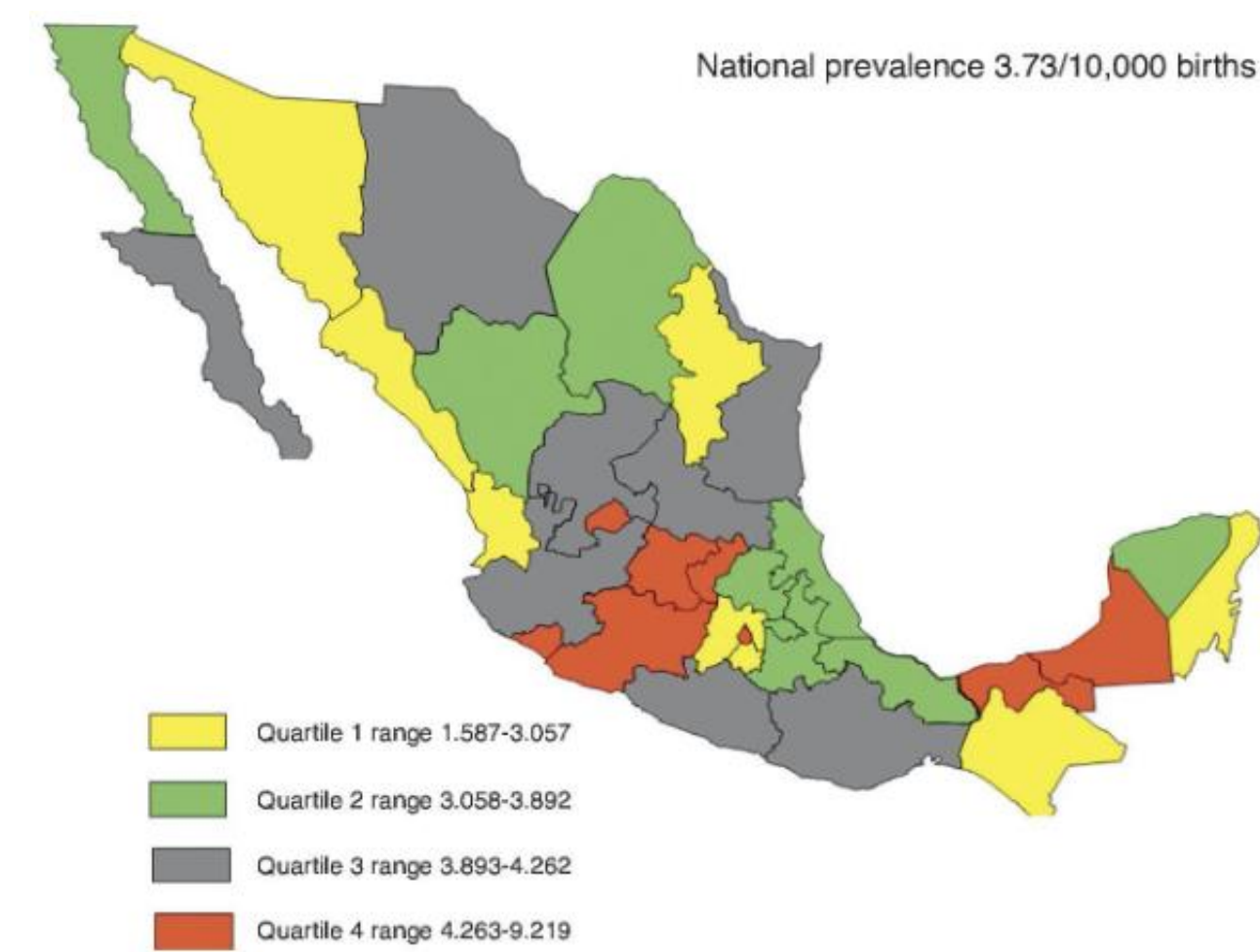
This project focuses on better understanding the prevalence and distribution of the Down syndrome birth rate in México. We used spatial statistics and linear regression to better understand whether Down syndrome births are geographically concentrated and what spatial processes are at play. Once we better understand where the concentration is, we can look at factors that may be influencing the spatial outputs. Once we better understand where the higher Down syndrome birth rates are, we can better understand where the resources need to be allocated. This work is completed through the Dr. Albert H. & Greta A. Bryan Summer Research Program in Mathematics at Simpson College.

2. PREVIOUS WORK

2.1 Estimates of the number of people with Down syndrome in the United States by age group from 1950-2010. [1]



2.3 Down syndrome live birth estimates in México from 2008-2011. [2]



2.4 Previous work takeaways:

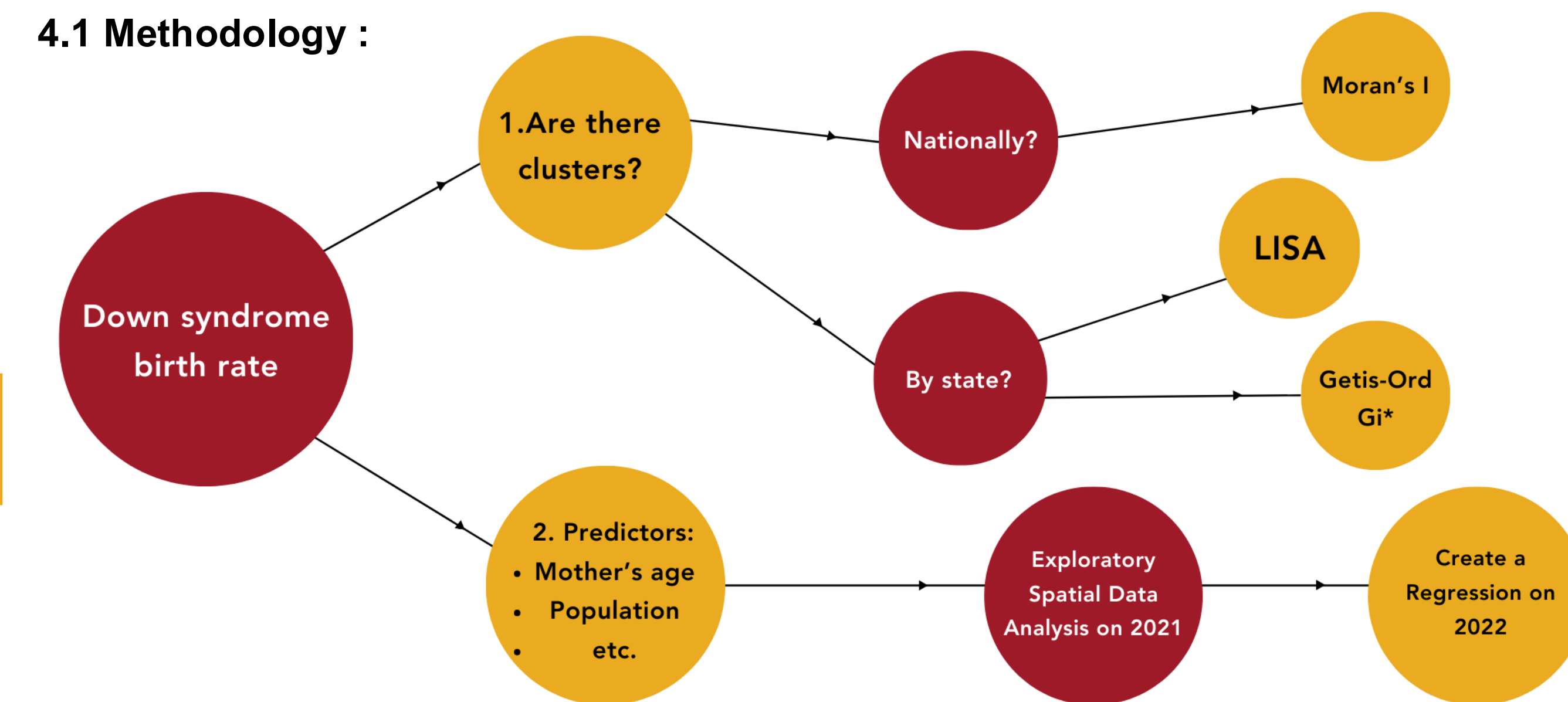
- Down syndrome is caused by a random error in cell division that results in the presence of an extra copy of chromosome 21
- Children with Down syndrome have the highest need in their first year of life
- The de Graaf et al. paper estimates the population of Down syndrome in the United States grouped by age
- Estimates of Down syndrome live births have been done at the state level in México from 2008 to 2011

3. RESEARCH OBJECTIVES

- Is the Down syndrome birth rate in México randomly distributed?
- What socioeconomic and environmental variables play a role in the Down syndrome birth rate distribution in México?

4. METHODS

4.1 Methodology :



4.2 Formulas:

National Level:
Moran's I

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=0}^n w_{ij}} \cdot \left[\frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x}) (x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

- n is the number of observations
- x_j are the values of the variables of interest at the locations i and j
- $\bar{x}$ is the mean of x_j
- w_{ij} is the weight that indicates the strength of interaction between locations i and j
- S is the standard deviation which measures the dispersion of the data points from the mean

State Level:
Getis-Ord Gi*

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} x_j - \bar{x} \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{[n \sum_{j=1}^n w_{ij}^2 - (\sum_{j=1}^n w_{ij})^2]}{n-1}}}$$

5. RESULTS TO OBJECTIVE 1

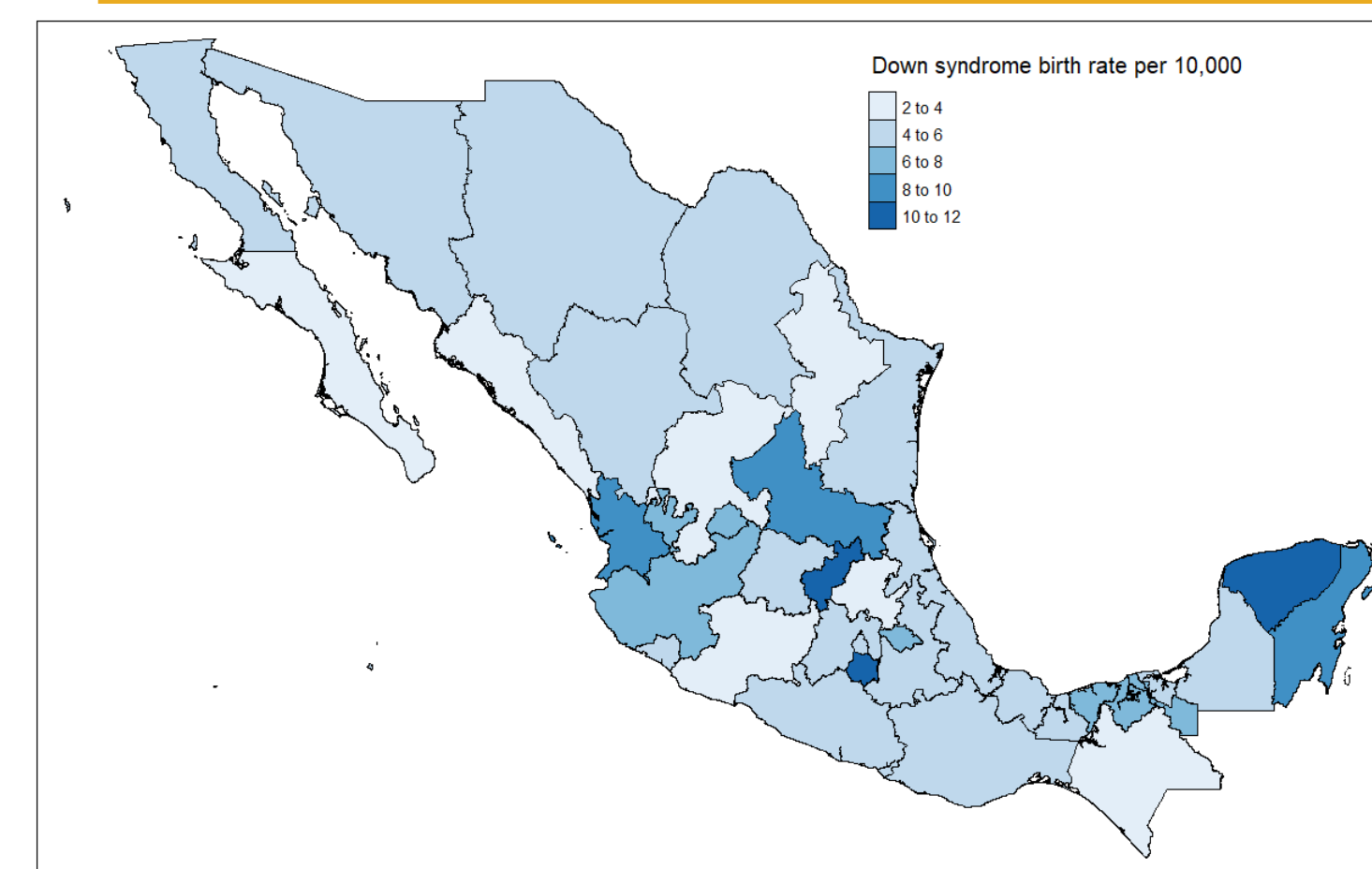


Figure 1: 2022 Down syndrome birth rates in México per 10,000 occurrences. 2018-2023 data were mapped, 2022 is representative of the patterns seen in the 6 years. The national average Down syndrome birth rate for 2022 is 5.77. The birth rate is higher in central México and the Yucatán Peninsula. [3]

Moran's I: -0.05
p-value: 0.56

The resulting Moran's I & p-value defines the distribution at the national level as random, meaning there are no definite clusters at this level.

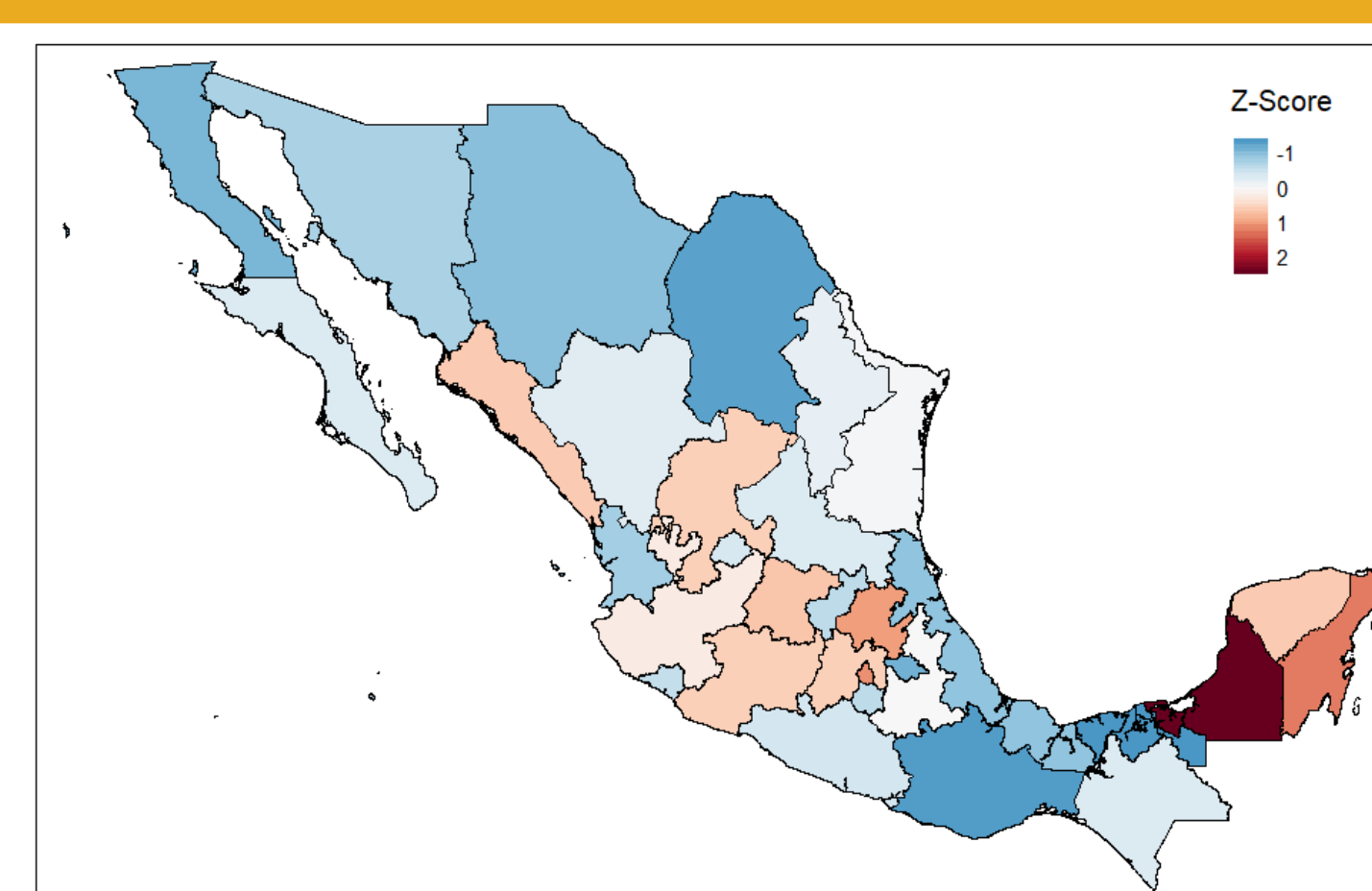


Figure 2: The Getis-Ord Gi* map on 2022 data. This confirms that there is a hot spot in the Yucatán Peninsula, making it an area of interest. [4]

6. RESULTS TO OBJECTIVE 2

2022 Predictors Output		
Predictors	Estimate	p-value
Intercept	-13.05	0.61
Mother's Mean Age	0.80	0.37
Population	-0.60	0.38
Percent of Catholics	0.04	0.33
Percent Uninsured	0.01	0.88
GDP per Capita	0.76	0.74
Percent of Maya Speakers Ages 5+	0.22	0.04
Multiple R ² : 0.2772		p-value: 0.1892

Table 1: These results show an unexpectedly high p-value for Mother's Age, as literature shows a large correlation between increasing age and Down syndrome birth rate. The most predictive variable is the Percent of Maya Speakers.[5]

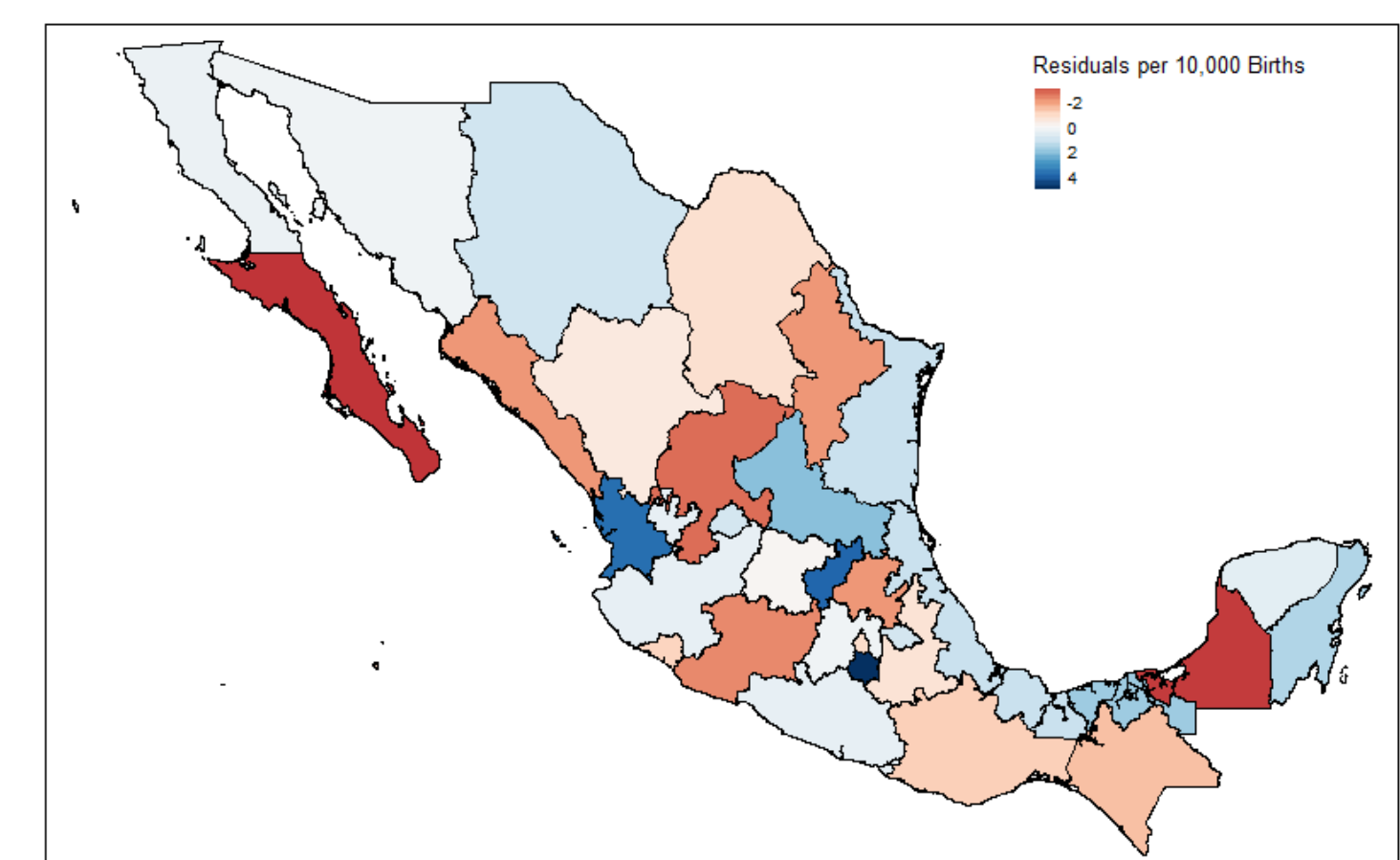


Figure 3: Residual output for predicting 2022. The map shows the overpredictions in red, and significantly so in Campeche, which is on the Yucatán Peninsula. The dark blue, or under-estimates are significant in Morelos, south of México City. Overall, the map captures spatial features of Down syndrome birth rates.

7. CONCLUSIONS & FUTURE WORK

7.1 Conclusions:

- The patterns of the Down syndrome birth rates in México from 2018-2023 are not random

- On the national level, there appears to be no clustering
- On the state level, there is a significant hot spot in the Yucatán Peninsula

7.2 Future Work:

- Better understand the Indigenous communities, especially the Yucatán Mayas and their culture
- Use a hierarchical Bayesian approach given the structure of our data (a Down syndrome birth is extremely rare)
- Model Down syndrome births at the individual level in México
- Use a Zero-inflated model at the municipality level to model the Down syndrome birth rate

8. REFERENCES

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